RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Wednesday, 7 October 2026, 8:54 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 8:54 AM
Time taken	25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Comparator function to sort integers in ascending order
int compare_asc(const void *a, const void *b) {
    return (*(int*)a - *(int*)b);
}

int main() {
    int n;

    // Read number of elements
    if (scanf("%d", &n) != 1) {
        return 1;
    }

    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the array elements
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 1;
        }
    }

    // O(N log N) Sorting using built-in Quicksort
    qsort(arr, n, sizeof(int), compare_asc);

    long long max_sum = 0;

    // Greedy step: Multiply elements with their corresponding index
    for (int i = 0; i < n; i++) {
        max_sum += (long long)arr[i] * i;
    }

    // Print the maximum array sum
    printf("%lld\n", max_sum);

    // Clean up memory
    free(arr);
```

```
45
46      return 0;
47  }
48
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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